

When is expensive perception worth paying for?

Jiaming Wei

MSc thesis · REALM workshop, EMNLP 2026

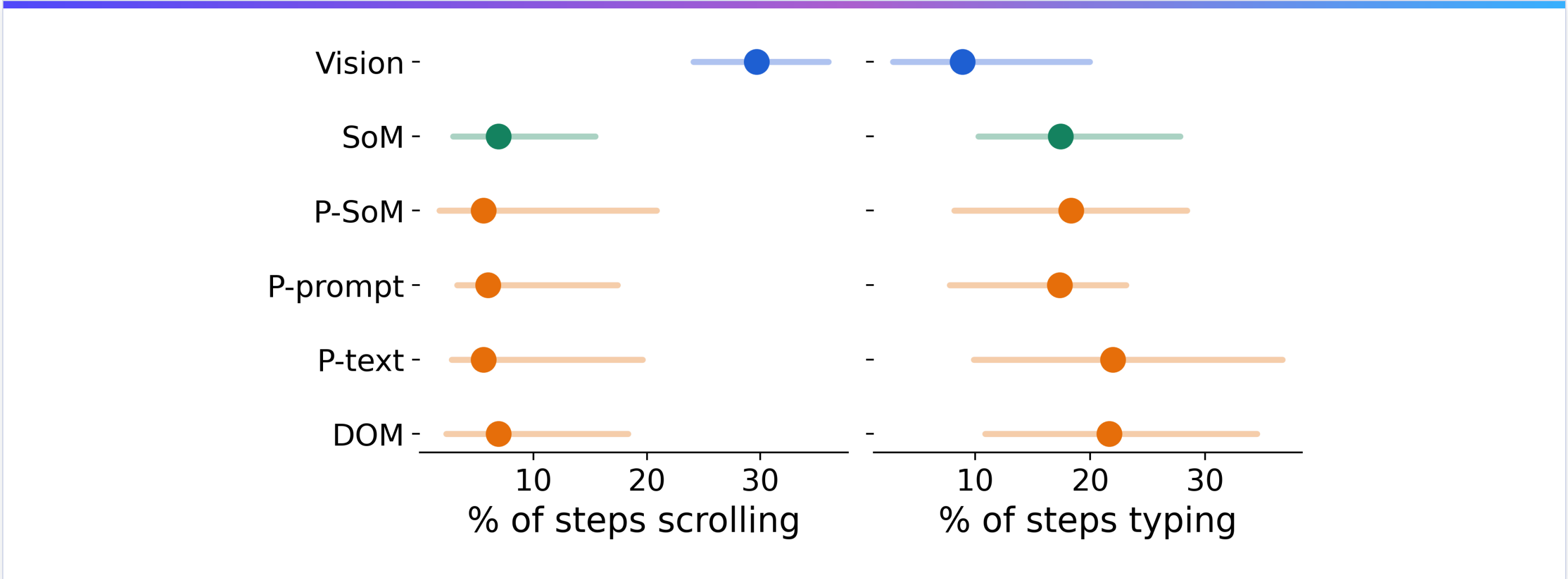
ONE RECORDED RUN PER VIEW · NOTHING ON THIS SCREEN IS LIVE

■ Same task, three ways of seeing it, three different bills.

IF YOU BUILD WEB AGENTS, TAKE AWAY

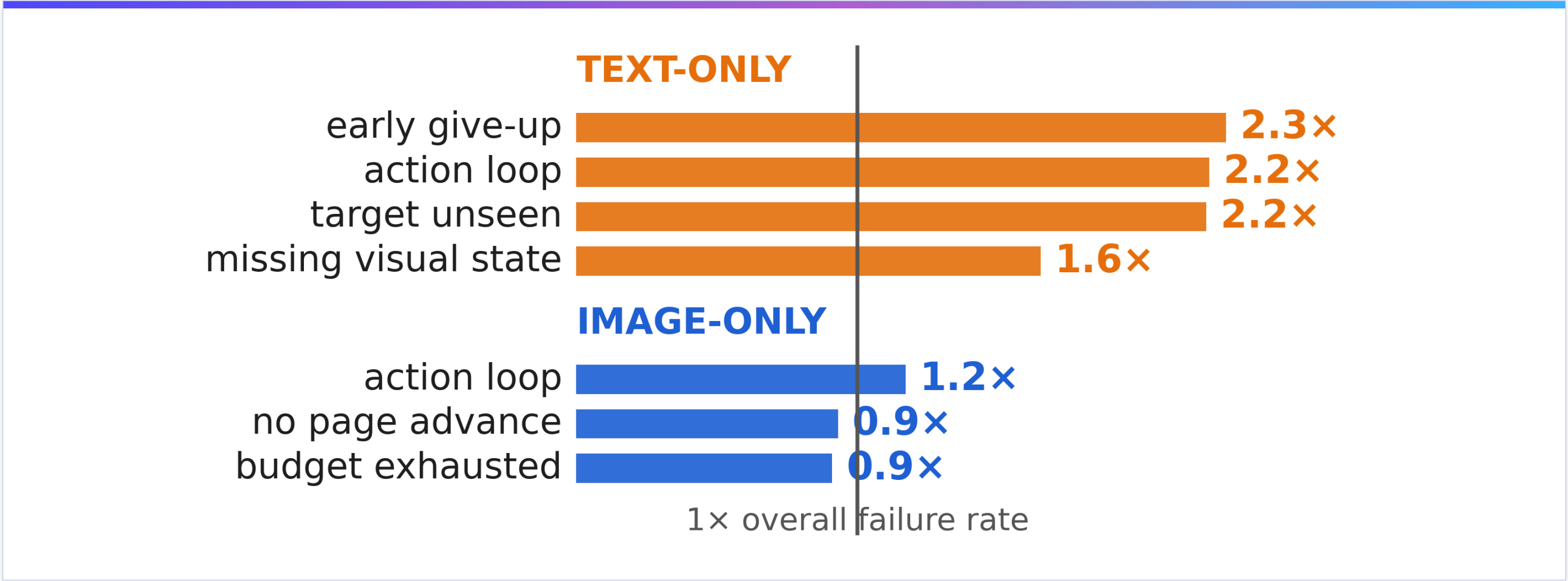
- 1 How an agent sees the page changes what it does, and how it fails.
- 2 Choosing the view per task would pay, in hindsight.
- 3 No learned choice gets there yet; a stronger agent is the test.

■ With only a screenshot, the agent scrolls far more and types far less.



Medians over 8 settings: image-only scrolls in **about 30%** of steps, the rest **about 6%**; it types about half as often

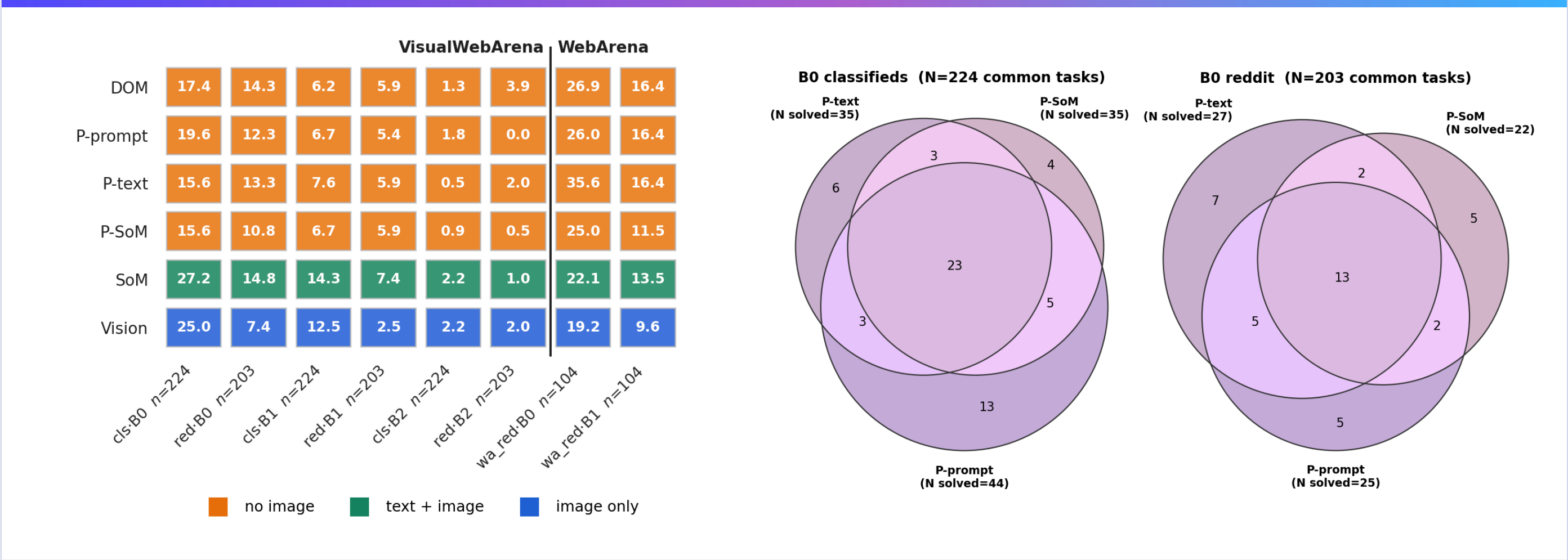
■ Text-only fails in ways you can name; image-only just never gets there.



Each side against its own failures elsewhere · text-only: several named failures over **2x** · image-only: none stands out · pattern from the six VisualWebArena settings

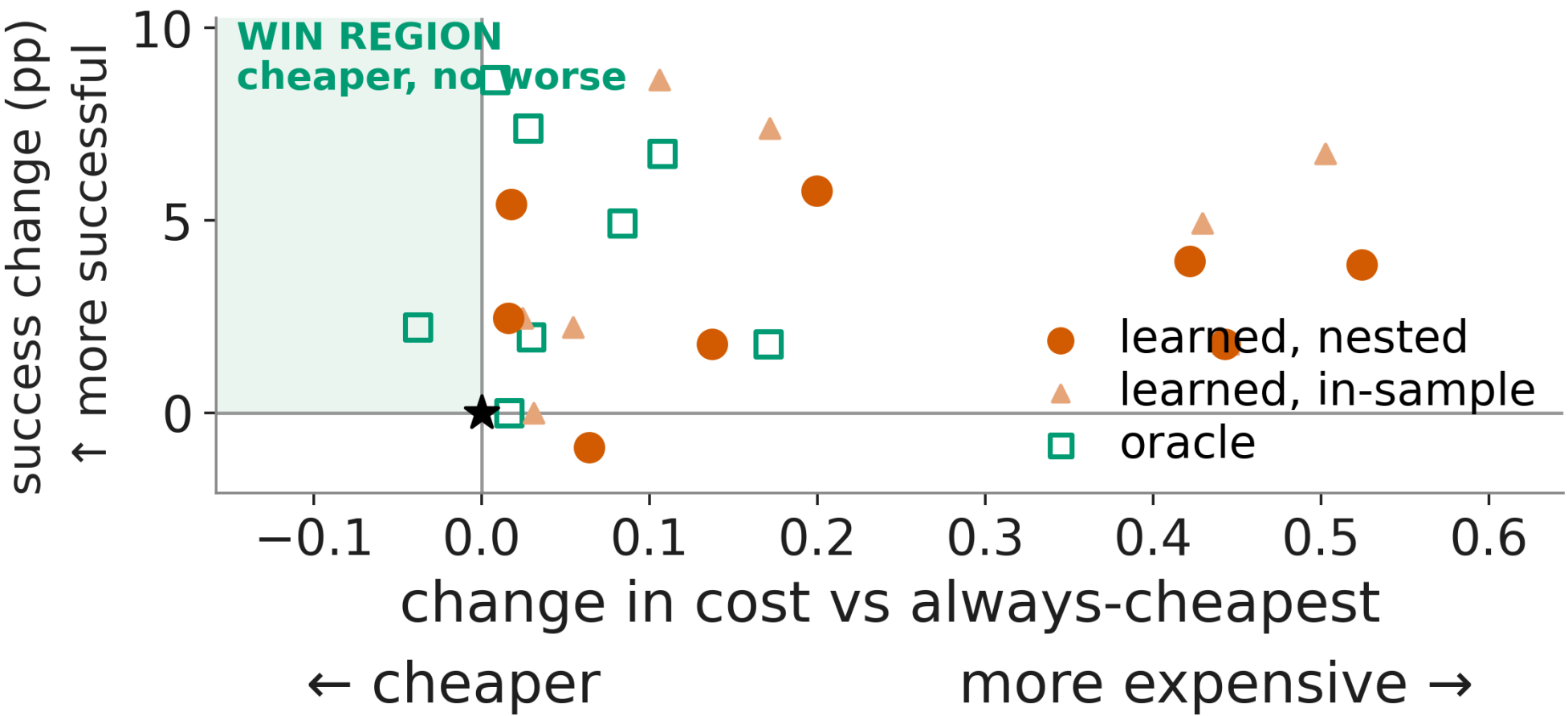
SO CHOOSE THE VIEW PER TASK? · NO VIEW WINS EVERYWHERE · THEY SOLVE DIFFERENT TASKS

■ Hindsight says choosing pays; a rerun says how much of that is noise.



Hindsight: **+3.45 to +16.35** in 100 vs the best single view · a rerun flips **10–14%**, buys **2.0–7.6**

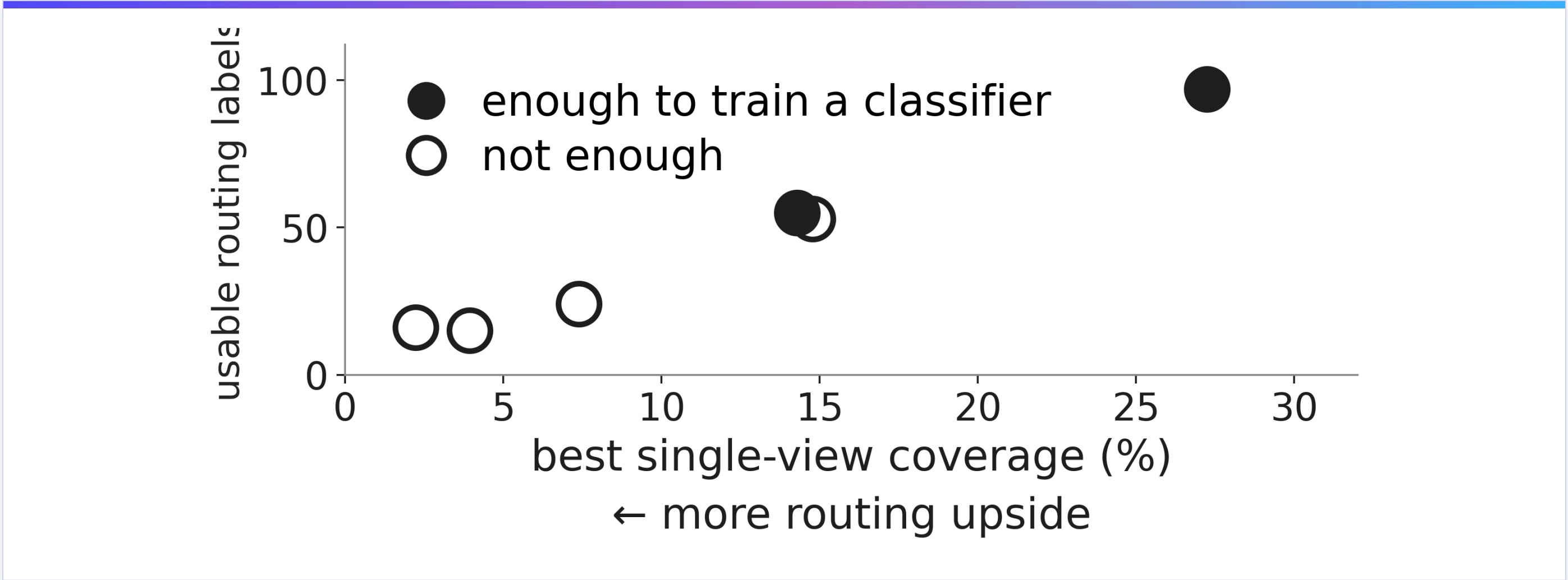
■ Learned routers buy success only by spending more.



Win region = cheaper and no worse. Learned: **0 of 8**. Perfect hindsight: **1 of 8**.

WHY NOT YET · A “WHICH VIEW” EXAMPLE EXISTS ONLY WHEN THE TASK IS SOLVED

■ More routing upside, less usable training signal.



Base success **2–36%** · **15–97** usable examples per setting · enough to train in **2 of the 6**

NOT YET

■ Improve the agent first, collect reliable examples, then learn when to look.

WHAT IT TAKES

- a stronger agent
- failing settings: at least **2.1–4.2×** more tasks
- reruns that agree

THREE ASKS

- ✓ type a task at the board
- ✓ which view do you use?
- ✓ lend us a stronger agent

Look, read, or both?
Hindsight says: choose.
Nothing we trained knows it yet.



Come to the board — try your own task, live in all three views.

Jiaming Wei · jiaming.wei.25@ucl.ac.uk

BACKUP – TASK 130, RECORDED STEPPING, 39 S